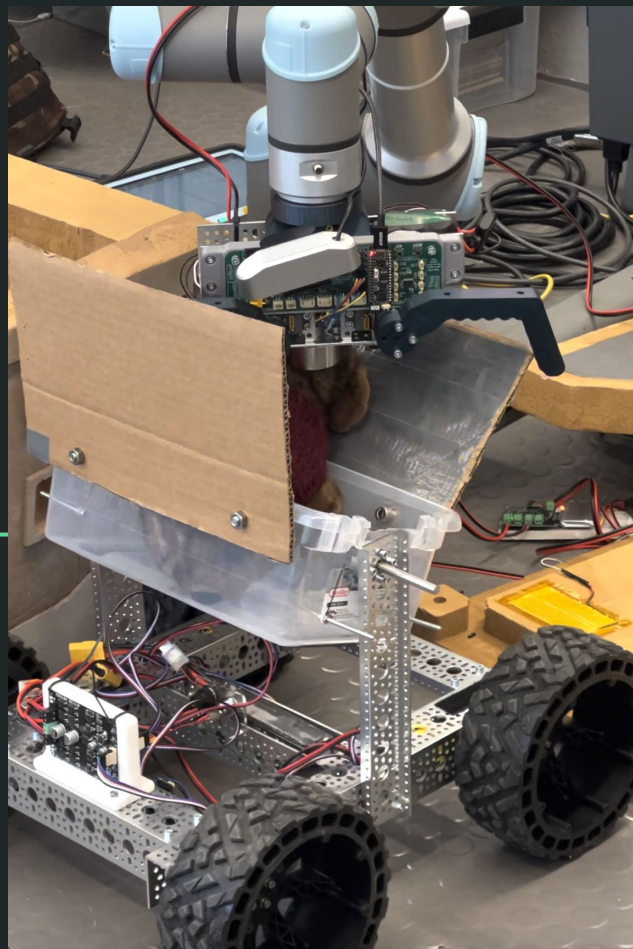
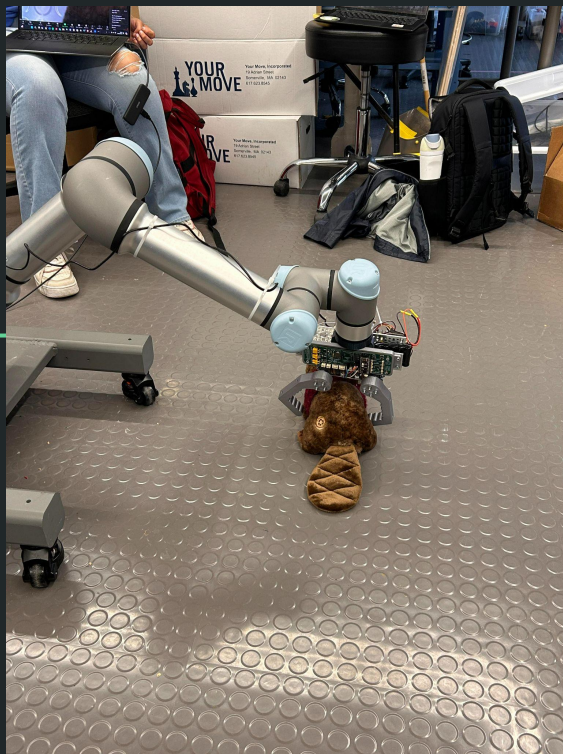
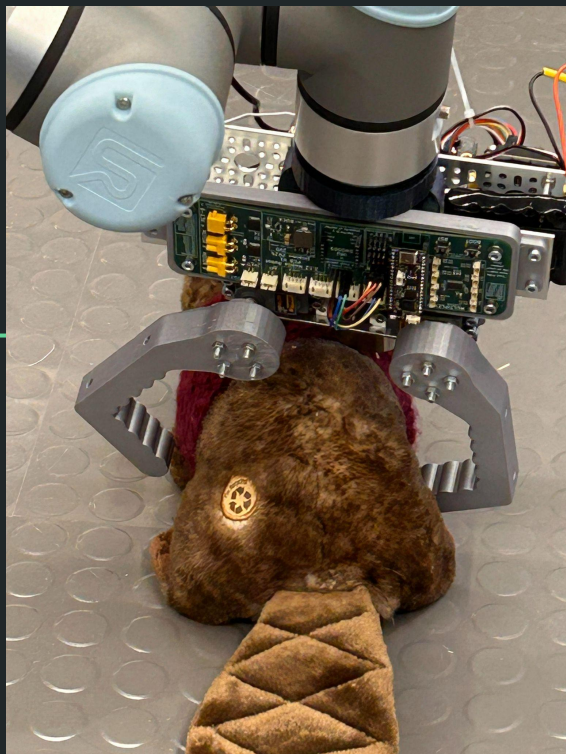


R3D5



Overview of Events

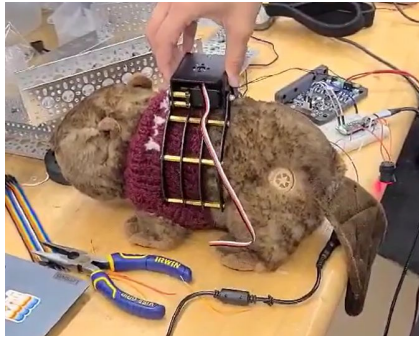
1. UR5 finds Tim's body heat with IR sensor
2. UR5 removes the building roof with electromagnet and picks up Tim with gripper
3. Mobile robot moves through course using the skycam to navigate
4. UR5 drops Tim in mobile robot crib

General Strategy

- UR5 moves manually with a controller.
 - Some automated repeated actions.
- Mobile robot moves manually through the course.
 - Large wheels to go over obstacles, swaying crib to limit rapid acceleration.

UR5 Gripper Design Process

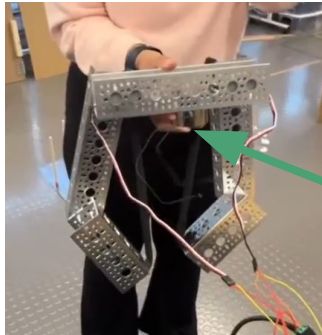
Goal was to create a gripper to pick Tim up but not crush him.



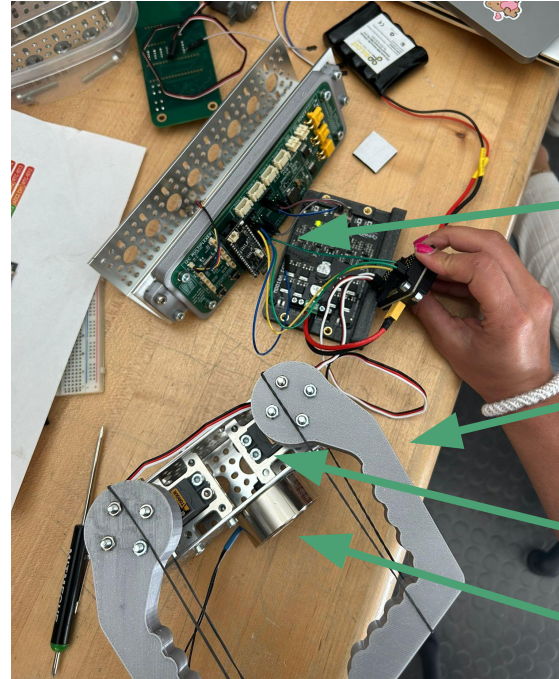
Amazon One
Servo Motor
Gripper



Needed more
customization



Two
servos



3D printed
UR5 mount

IR sensor

3D
printed
Claws

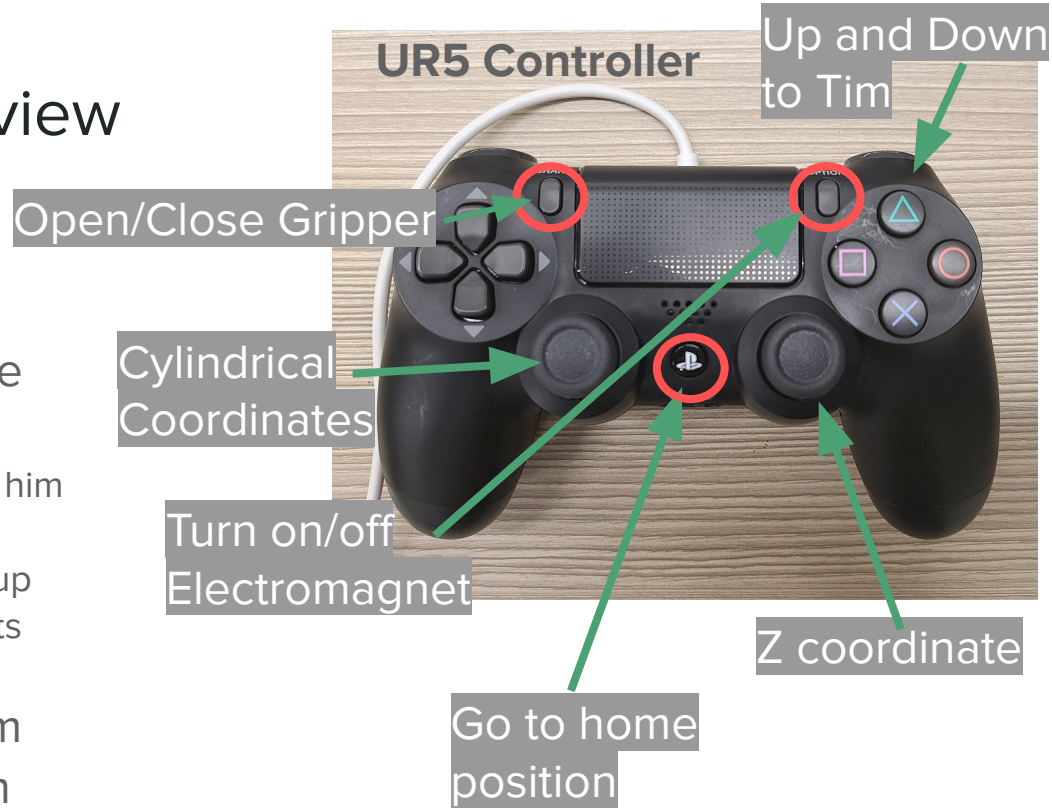
webcam

Electromagnet



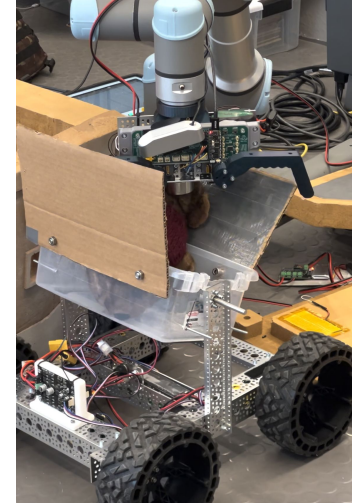
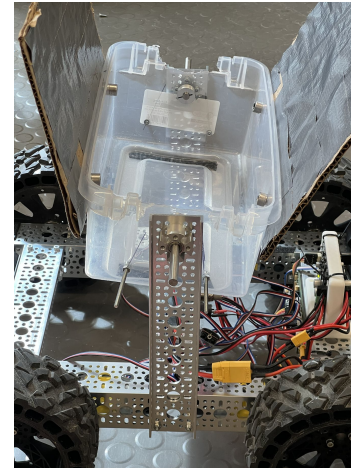
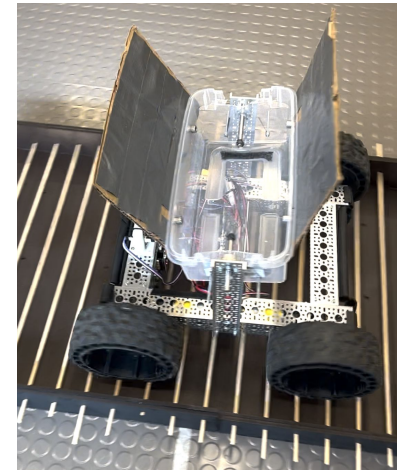
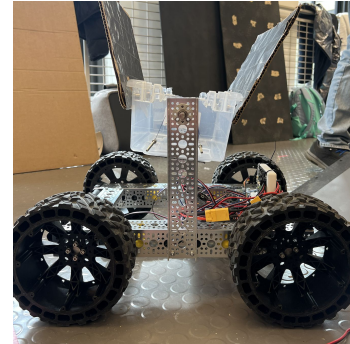
Code and Controller Overview

- Created autonomous series of movements mapped to buttons and keys- automatic functions we have:
 - Moves down to Tim's height to pick him up then back up.
 - Goes to specified positions to pick up the roof. Attaches to the roof and lifts up.
- Implemented code to predict arm singularities and prevent the arm from locking.



Mobile Robot Design Process - Structure

- **Chassis size:** reduced the width of the mobile robot.
- **Wheels:** opted for Wasteland Wheels (192 mm diameter and 72 mm width). Greatly improved speed, mobility, and traction.
- **Crib structure:** dowel swing system and funnel for the crib (to avoid tilting or dropping Tim).



Mobile Robot Design Process - Control

- Strategy: Quickly go through the treacherous path
- **Custom controller:**
 - **2 joysticks for velocity input** - one for forward/backward and one for turning.
 - **Low pass filter on joystick readings:** moving average filter to smooth the joystick movements.
 - **Switch for speed setting:** switch between high- and low-speed mode depending on situation.
- Speed limits defined in code

